# 2.4 Free Fall

Introduction. This handout is printed in two columns. Read the left column from top to bottom before moving to the right column, and finish with the summary at the end of the page.

Paragraph A. An object in free fall moves only under the pull of gravity. Near the ground every object speeds up at the same rate, whatever its mass, once air resistance is small enough to ignore.

Paragraph B. The velocity after a time $t$ is $v = gt$ for an object dropped from rest, so the velocity grows steadily while the object falls.

Paragraph C. The distance fallen grows faster than the time, because the object is moving faster during every later second than it was during the second before.

Paragraph D. A simple experiment drops a ball past two light gates and records the time between them. Repeating the drop several times reduces random error.

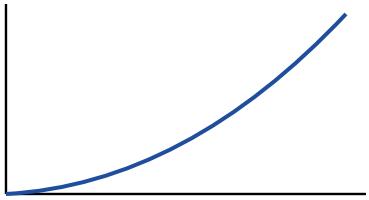

Figure 2.4 Distance fallen against time for a dropped ball.

Paragraph E. The curve in the figure bends upwards, which is what a steadily increasing speed looks like on a graph of distance against time.

Summary. Free fall is motion with constant acceleration, so the same equations describe a dropped ball, a falling stone and a cart rolling down a smooth ramp.